

Exposé

Provadis School of International Management and Technology

**Comparing CPython 3.14 JIT, PyPy, and Cython for
Smith-Waterman Sequence Alignment**

Performance Evaluation and a Robust Cross-Runtime Testing Suite

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Glossary

Term	Description
Smith-Waterman alignment	A dynamic-programming algorithm for local sequence alignment that computes the highest-scoring matching subsequences between two input sequences.
local alignment	An alignment strategy that searches for the best matching subsections of two sequences instead of forcing an end-to-end alignment.
warm-up phase	The initial execution period in which a JIT-enabled runtime collects profiling information and may compile hot paths before reaching steady-state performance.
benchmark harness	The controlled code and tooling used to execute reproducible performance measurements across implementations and runtimes.
differential testing	A testing strategy that compares the outputs of multiple implementations against the same inputs to detect behavioral deviations.

Abbreviations

Abbreviation	Meaning
JIT	Just-in-Time
DP	Dynamic Programming
SW	Smith-Waterman
CI	Continuous Integration

Exposé

1. Problem Statement

2. Objectives

3. Methodology

3.1. Testing Suite Design

4. Expected Contribution

5. Planned Structure

The paper will likely be structured as follows:

- Introduction
- Research Question and Motivation
- Methodology
- Results
- Discussion
- Conclusion and Future Work
- References
- AI Declaration
- Declaration of Authorship

References

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